Save the Human (in the Loop): When Does Human Oversight of AI Actually Reduce Risk?

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Abstract: Human oversight is the standard safeguard for AI systems in regulated industries and is widely assumed to reduce risk by default. We test that assumption on a simulated fraud review built on the public FiFAR dataset. Three ways of deciding the 4,457 alerts of one test month are compared: full automation by a gradient-boosted classifier, naive review of every alert within a fixed review capacity, and analyst-aware deferral, which sends each alert to whichever decision-maker is estimated most likely to decide it correctly, a simulated reviewer or the classifier. Each review policy is run with three teams of simulated reviewers: a standard team, a team that mostly follows the model, and a team biased against applicants aged 50 or over. Risk is measured on three axes that are never combined: misclassification cost, frauds accepted, and the size of the age gap in wrongful rejections. Every verdict comes from decision rules fixed in the study's design, with paired bootstrap ranges adjusted for multiple comparisons. The hypothesis that deferral achieves lower risk than both alternatives is refuted on all three measures. On misclassification cost, analyst-aware deferral beats full automation for every team at every realistic capacity at which it is judged, but naive review never falls below automation either, and deferral is not shown to beat naive review in all three comparisons. On frauds accepted and on the age gap, deferral fails to beat full automation. The biased team "trusts" that the classifier rejects older applicants correctly, thus such cases are always deferred to the classifier. On each measure the benefit of human review is smaller and more conditional than assumed.

This study was made for research only.

I. Introduction

A. Problem and Objective

Practice suggests that human oversight does not guarantee risk reduction: reviewers "are given more alerts than they can handle, review cases the model already gets right, or defer to the model score instead of challenging it", and "what is missing is an operational answer to the question of when oversight genuinely helps" [4]. The project's objective is to "define what 'appropriately applied' human-in-the-loop, hereafter HITL, means in measurable terms, and test that definition empirically on a fraud-review use case" [4]. To make that objective measurable, this study proposes four criteria for appropriate HITL; section IV-E states them and how each is treated.

Every result table and graph in this report is produced by the code of the repository *https://github.com/zsipka/SaveHITL-pub* [5]. Its notebook, `savehitl_results_final.ipynb`, recomputes the whole chain from the public FiFAR dataset, starting from the dataset's alerts file and the framework's Classifier *h*. The repository also holds the short notebook, `savehitl_results.ipynb`, and the long notebook, `savehitl_results_long.ipynb`: the short and the long version of the notebook. Every value in this report's result tables and graphs is either shown in the notebook or written by its run to a table file in its folder `tables/`. The tables and graphs the notebook produces are computed from FiFAR (Alves et al., figshare, 2025, *https://doi.org/10.6084/m9.figshare.28351172*, CC BY 4.0, *https://creativecommons.org/licenses/by/4.0/*). They are derived from FiFAR by this project, not part of it, and FiFAR's creators have not reviewed or endorsed them.

B. The Hypothesis and How It Is Read

The project outline states: "Across realistic review capacity levels, calibrated deferral achieves lower risk than both full automation and naive review-everything HITL. Naive HITL can be worse than full automation when review capacity is exceeded or when reviewers largely agree with the model" [4]. The study replaces calibrated deferral with analyst-aware deferral, one of the outline's aspirational extensions, as it better illustrates the objective.

The hypothesis therefore has two parts. **First**, across realistic review capacity, analyst-aware deferral achieves lower risk than both full automation and naive review of every alert. **Second**, naive review can be worse than full automation when capacity is exceeded or when reviewers largely agree with the model.

How the study reads the hypothesis, and when it is refuted, is fixed in the policy designs [10], [11] and set out in section II-G. Every combination is reported, including those unfavourable to the hypothesis.

C. What the Study Compares, and What It Uses

This study compares three policies for deciding fraud alerts, each specified in its own policy design [9]–[11], and draws its conclusions from those designs. It uses the FiFAR dataset [2], and it uses the OpenL2D framework [3] of Alves et al. [1] to generate the simulated reviewers, to reuse models and to train the router by the framework's own procedure. The study's guiding principle is that its mission is not to build the best model, but to assess and compare HITL options. The reused models and the simulated reviewers are therefore used as mechanisms that generate data. What matters is that every policy uses the same cases, models, settings and rules, so that differences come only from the routing rule compared. Section II-H states, part by part, what is reused unchanged, what is adapted, and what is this project's own (Table 1).

D. Organisation of the Report

Section II describes the data, models, reviewers, teams and policies, the measures, and the decision rules fixed in the design, and ends with what the study reuses and what is its own. Section III gives the deciding results and the verdict on the hypothesis. Section IV gives the descriptive results, which decide nothing. Section V sets out how the study would apply in real use, and the further aspects that commercial use would need worked out in law. Section VI discusses departures from the project outline, the limitations and the risks; section VII concludes. The appendices hold the detailed deciding results, the data by age group and the descriptive verdicts across the capacity grid.

II. Data and Set-up

A. Data and Scope

FiFAR is built on the Bank Account Fraud dataset: 1,000,000 generated bank account applications over eight months, each labelled as fraud or legitimate. The dataset's Alert Model scores every application from month 3 on and flags those above a cut-off, set so that 5% of the legitimate applications of month 3 are flagged. The flagged applications are the alerts, and only alerts reach Classifier h or a reviewer. Results are measured on month 7, the test month, whose alerts are the cases the policies decide. Months are numbered 0 to 7, as in the data files.

Frauds that the Alert Model does not flag are ignored in this study. They contribute to the effectiveness of the Alert Model, which is not the subject of the study. Frauds can be controlled by the alert threshold in practice. We use a fixed threshold at a 5% false positive rate.

Which months are used for what (Fig. 1). The Alert Model learns on months 0 to 2; its settings and cut-off are chosen on month 3; it scores months 3 to 7. Classifier h learns on the alerts of months 3 to 5, is selected on month 6 and decides the alerts of month 7. The router's models learn on months 3 to 5, have their settings chosen on month 6, and score the alerts of month 7. The 50 reviewers come last: they are not machine-learning models but simulated reviewers, whose baseline settings are fitted on the alerts of months 3 to 6 and who hold a decision on every alert of months 3 to 7. Month 7 is used only for results: nothing is learned, chosen or set on it.

Every FPR in this report, except in Table 4, is taken among the alerts, not over all applications. These FPRs therefore appear much higher than a bank's rejection rate across the full population of applications. The reason is that the study assesses how effective human review is, and its starting point is the alerts: applications already flagged as suspicious.

Terminology clarification. A confirmed alert means the application is rejected, and a dismissed alert means it is accepted; each subsection uses one of the two forms.

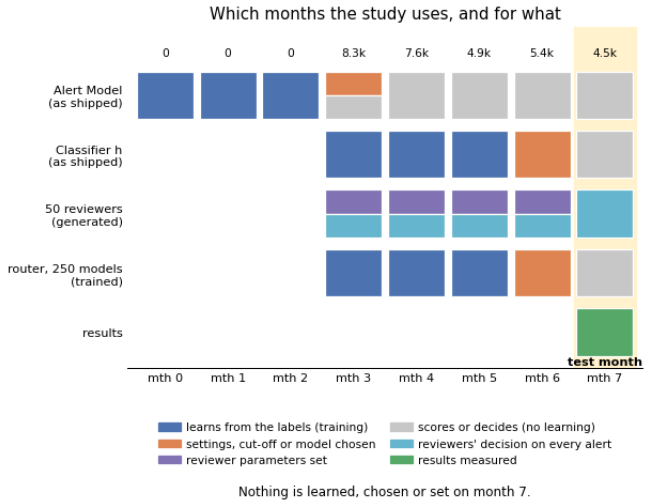


Fig. 1. *Which months each model and the reviewers learn from or are set on, and which month gives the results.*

B. Models

The study does not retrain the framework's Alert Model or Classifier h, for efficiency. Two are reused as shipped. The **Alert Model** scores every application; its score above the cut-off makes an application an alert, and it decides nothing in any policy. **Classifier *h*** decides each alert on its own: it confirms the alert when its score is 0.5 or more. It was trained by minimising the weighted cross-entropy loss $L = -\sum_i w_i\ [\, y_i \log p_i + (1-y_i)\log(1-p_i)\,]$, where $y_i = 1$ for a fraud and 0 for a legitimate case, p_i is Classifier h's score for case i, and $w_i = 1$ for frauds and 0.057 for legitimate cases.

The study fits and tunes the **router**, the framework's OvA Classifiers: one classifier per reviewer, estimating the probability that the reviewer decides a given alert correctly. There are 250, one per reviewer for each of the five team versions and five training shuffles, trained on the decisions of this study's reviewers by the framework's own procedure, including its settings search of 120 trials per model. That search was run in advance; by default the notebook refits the models from the settings it chose.

All three models are LightGBM models, a library for gradient-boosted decision trees, as the framework sets them up [3]. The settings searched are the number of trees, their depth, the number of leaves, the minimum number of cases per leaf, the learning rate and two regularisation terms [1, pp. 7, 19, 23].

C. Reviewers and Teams

How the Synthetic Reviewers are generated. The framework's Synthetic Experts are not trained models. Each takes an alert's true label and flips it with a probability that depends on the alert's features [1, p. 3]; machine learning calls this instance-dependent label noise [1, p. 1]. For expert j and case i, the generator gives the chance of wrongly rejecting a legitimate application and of wrongly accepting a fraudulent one by the following formula, formula (1) of [1, p. 3]:

$$\mathbb{P}(m_{j,i} = 1 \mid y_i = 0, \boldsymbol{x}_i) = \sigma\left(\beta_0 + \alpha \frac{\boldsymbol{w} \cdot \bar{\boldsymbol{x}}_i + w_M M(\boldsymbol{x}_i)}{\sqrt{\|\boldsymbol{w}\|^2 + w_M^2}}\right),$$

$$\mathbb{P}(m_{j,i} = 0 \mid y_i = 1, \boldsymbol{x}_i) = \sigma\left(\beta_1 - \alpha \frac{\boldsymbol{w} \cdot \bar{\boldsymbol{x}}_i + w_M M(\boldsymbol{x}_i)}{\sqrt{\|\boldsymbol{w}\|^2 + w_M^2}}\right).$$

Here $m_{j,i}$ is the expert's decision (1 rejects as fraud) and y_i the true label (1 fraudulent).

This study's reviewers. The 50 reviewers are generated with the dataset's own generator, with two changes to its settings: no reviewer gives any weight to the applicant's age, and each reviewer's baseline settings are set so that, on the alerts of months 3 to 6, its FPR (legitimate applications wrongly rejected ÷ all legitimate) and FNR (frauds wrongly accepted ÷ all frauds) match targets drawn for it. The changes limit age bias to a single team, so that the teams' impact can be compared, and fit each reviewer's baseline settings on four months of alerts instead of one. Each reviewer is built to a target cost per case that the author, from domain experience, considers realistic for experienced subject matter experts. Each reviewer's decision on every alert is generated once and replayed by every policy.

Teams. A team has ten reviewers and is run in five versions: five disjoint sets of ten, drawn at random from the 50 with a fixed seed, so each reviewer is in exactly one version. Within a version, the three teams are made of the same ten reviewers and differ only in how they decide a reviewed alert:

- **Team A, standard mixed team (the control):** members' own decision stands (behaviour diagram: policy 2, policy 3).
- **Team B, agrees with the model:** members follow the model in 4 of 5 reviewed cases, picked at random; otherwise their own decision stands (behaviour diagram: policy 2, policy 3).
- **Team C, biased:** members reject older applicants (aged 50 or over) whenever the model would (behaviour diagram: policy 2, policy 3).

The precise behaviour of each team is fixed by the teams' decision trees in the designs: all three for naive review in policy 2, section 2.1 [10], and for analyst-aware deferral in policy 3, section 2.1 [11]. Capacity is not a property of a team: every team is run at every capacity.

D. The Three Policies

Each policy is specified in its design: policy 1 in [9], policy 2 in [10] and policy 3 in [11].

Policy 1, full automation. Classifier *h* decides every alert with its built-in rule; capacity and team play no role. For reference only, a second automated option is shown beside it: reject every application.

Policy 2, naive review. Every alert is meant for review. A team can review only a share of the alerts, the capacity. For each random order, the alerts are taken in one random order and those within capacity are reviewed, dealt to the team's ten reviewers. Every alert beyond capacity is confirmed without review, i.e. the application is rejected. Overload means that review capacity is below 1, so the alerts the reviewers cannot handle are confirmed without review and those applications are rejected. The policies differ in who decides, not in which order alerts come, for better comparison.

Policy 3, analyst-aware deferral. The alerts are taken in the same random order. For each alert the router estimates how likely each of the team's ten reviewers is to decide it correctly, and Classifier *h*'s estimate is its confidence in its own decision. An alert goes to a reviewer only where that reviewer's estimate is strictly higher than Classifier *h*'s and capacity remains. The ten reviewers share one pool the size of the capacity; Classifier *h* has no limit.

E. What Is Varied

- **Capacity:** 0 to 1 in steps of 0.05, plus 10/11: 22 values. 10/11 is the benchmark's own setting: its reviewers receive ten of every eleven alerts, and Classifier *h* decides the eleventh. The **realistic range**, this study's choice, is the nine points 0.65, 0.70, 0.75, 0.80, 0.85, 0.90, 10/11, 0.95 and 1. This study takes 0.75 as the capacity stress boundary for teams B and C, where they are compared with team A at the same capacity.
- **Cost ratio** λ, the cost of wrongly rejecting a legitimate applicant divided by the cost of missing a fraud: 0.057, the ratio Classifier *h* was trained for, on which every deciding result rests; and 0.035, 0.0537 and 0.171, descriptive. The decisions do not change with λ; only misclassification cost does. A cost without a stated ratio is at 0.057.
- **Random orders:** five. A random order fixes the review order, team B's deference draws and, in policy 3, the router's training shuffle.
- **Team versions:** five.

Every combination of team, version, random order, capacity and cost ratio is run and reported; the full sweep is in the table files the notebook writes when it is run, `tables/P2-T5_full_sweep.csv` and `tables/P3-T5_full_sweep.csv` [5].

F. Measures, and How a Comparison Is Judged

Risk is measured on three axes, never combined; on each, lower is better:

- **Misclassification cost** $= \text{FN} + \lambda \cdot \text{FP}$, summed over the alerts, where FN counts frauds accepted and FP legitimate applicants rejected, in units of one missed fraud.
- **Regulatory exposure:** the number of frauds accepted, and the FNR among the test-month alerts. Verdicts are on the count.
- **Fairness exposure:** the age gap $= \text{FPR}_{50+} - \text{FPR}_{<50}$, aged 50 or over being the framework's declared protected group. Verdicts are on its size, |age gap|.

Differences are always the policy named first minus the one it is compared with, on the same alerts; below zero favours the policy named first.

Ranges come from a paired case bootstrap: the 4,457 test cases are re-drawn 50,000 times, both sides of a comparison are scored on the same re-drawn cases, and the range holds the middle of their differences, widened

for the number of deciding results. One index matrix of re-draws, drawn once from a fixed seed, is shared by all policies. Per-case results are averaged over the five random orders before re-drawing.

To describe in pseudo code terms:

```
MC(t, r, m) = misclassification cost of team t,
    version r, on redrawn month m
For each redrawn month m (1 to 50,000) and version r
    (1 to 5):
  Diff[r][m]  = MC(B, r, m) − MC(A, r, m)
  Diff_avg[m] = average of Diff[1][m] … Diff[5][m]

Range(X) = the 50,000 values of X without the top
    and bottom c%
Below(X) = every value in Range(X) < 0
Above(X) = every value in Range(X) > 0

if Below(Diff[r]) for all five r  => B beats A
elif Above(Diff[r]) for all five r  => B does worse
    than A
elif Below(Diff_avg)    => B beats A
elif Above(Diff_avg)    => B does worse than A
else                    => no difference shown
```

```
c = 5% error budget split across the deciding
    results, halved for each end:
    • policy 2: 61 results, c = 0.041%;
    • policy 3: 66 results, c ≈ 0.038%;
    • all other results: c = 2.5%
```

G. Decision Rules, Fixed in the Design

- **First refutation condition, "deferral fails to outperform automation at any capacity level":** if analyst-aware deferral fails to beat full automation in at least one of its deciding results against it (Table 2) on a measure, the hypothesis is refuted on that measure.
- **Second refutation condition, "naive HITL never falls below automation":** if no team under naive review does worse than full automation in any of its deciding results against it (Table 2; team A in Table A1) on a measure, the hypothesis is refuted on that measure. Every "does worse" counts, including one carried only by the alerts beyond capacity, which is labelled as a consequence of the overflow rule.
- **The second part, further rules.** *Capacity exceeded:* supported on a measure if team C does worse than full automation at one or more of the eight realistic capacities below 1. *Reviewers agree with the model:* supported on a measure only through team B's comparison with full automation, each "does worse" labelled by its cause. *Bias at equal capacity:* whether team C's rule costs or saves misclassification cost at 0.75 against the same reviewers without it.
- **The first part against naive review:** supported on a measure only if analyst-aware deferral beats naive review in all three comparisons (Fig. 3).

Away from 0.057. Neither Classifier *h*, the reviewers nor the router is re-tuned for the other cost ratios. Results there are descriptive and are shown in both directions in the notebook (section 4.9), and every "beats" there holds only "against an automation baseline not re-optimised for this cost ratio"; no conclusion holding across cost ratios is drawn.

H. What the Study Reuses, and What Is Its Own

Part	Reused unchanged	Adapted	This project's own
Cases	The alerts of the dataset's Alert Model at its 5% false positive rate cut-off; the test month, 4,457 alerts	—	Frauds the Alert Model does not flag are left out
Automated decision-maker	Classifier *h* as shipped, with its built-in rule (a score of 0.5 or more confirms the alert)	—	Its role as policy 1 and as one decision-maker in policy 3
Simulated reviewers	The framework's generator, configuration and seeds; the stored target costs and weights	Reviewers regenerated with no weight on age and each reviewer's baseline settings fitted on four months	—
Review teams	Ten reviewers per team	—	Five disjoint teams of ten drawn at random from the 50 reviewers with a fixed seed; teams A, B and C as three rules for the final decision
Router of analyst-aware deferral	The OvA method (one classifier per reviewer), its training procedure including its settings search (120 trials per model, lowest weighted log-loss), case weights and features; the framework's script that builds the training files	The OvA Classifiers trained on this study's reviewers, on training files built for this study's teams by the framework's script with one adaptation; the routing rule amended: Classifier *h* has no capacity limit, the reviewers share one pool, ties are resolved by fixed steps, and routing aims at each team's final decision	—
Cost ratio	0.057, the ratio Classifier *h* was trained for	—	0.035, 0.0537 and 0.171
Review capacity	10/11, the share the benchmark gives its own reviewers, as one grid point	—	The sweep from 0 to 1 and the realistic range 0.65 to 1

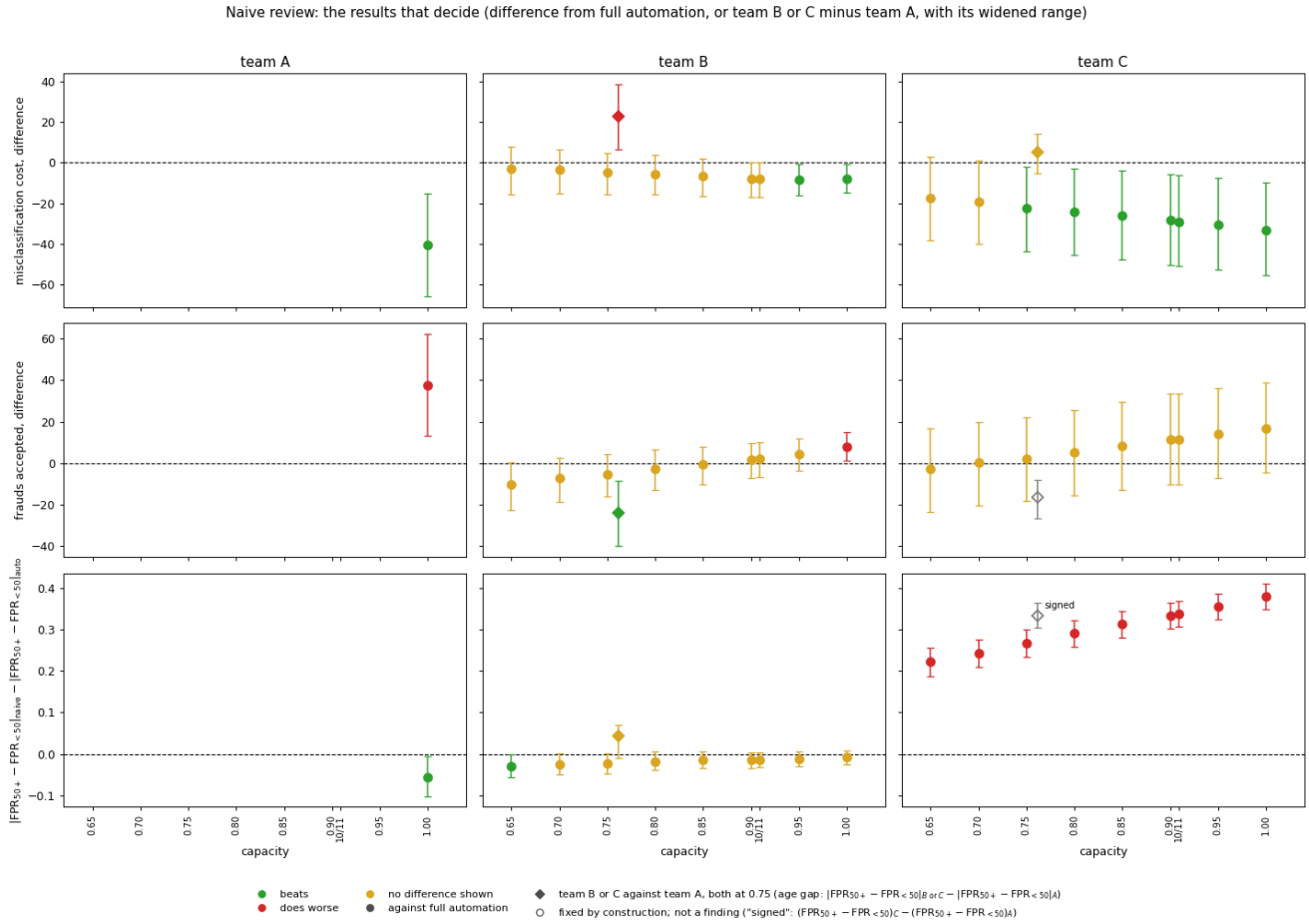
Table 1. *What is reused from the FiFAR dataset and the OpenL2D framework, and what is this project's own (the policy designs [9]–[11], section 3, "What is used")*

III. Results on the Hypothesis

Full automation, the baseline, has a misclassification cost of 204.05, accepts 40 of the 711 frauds among the alerts (FNR 5.6%), and has an age gap of 0.0696 (FPR 0.8166 at 50 or over, 0.7470 under 50; notebook, section 4.2, and Table 3). All figures below are means of the five team versions, each first averaged over the five random orders. The numerical deciding results are in Appendix A.

A. Naive Review against Full Automation

On misclassification cost (Fig. 2), naive review beats full automation with team A at full capacity (−40.76), with team B only at the highest realistic capacities, and with team C except at the lowest realistic capacities.



FPR among the alerts: the share of legitimate applicants in the alerts whose application is rejected (wrongly rejected ÷ all legitimate), in the age group shown (50 or over; under 50). |…| is the size of the gap. Below zero: the first policy's gap is closer to zero.

Fig. 2. *Naive review: every deciding result, one row per measure (for the age gap, its size), one column per team, the realistic capacities 0.65 to 1 across; difference from full automation with its adjusted range, coloured by verdict. In the panels of teams B and C, the diamond marker shows the team against team A, both at 0.75. Capacities below 0.65 are not judged and are shown as part of the notebook run. Graph: notebook, section 3.1, item P2-F10.*

On frauds accepted, it does worse with team A at full capacity (+37.52) and with team B only at full capacity, and shows no difference with team C.

On the age gap, it beats with team A, with team B only at the lowest realistic capacity, and does worse with team C at every realistic capacity. Elsewhere in the realistic range, team B shows no difference.

Within naive review at 0.75, team B does worse than team A on misclassification cost and beats it on frauds accepted; team C shows no difference from team A on misclassification cost, and its two other results follow from its rule.

B. Analyst-Aware Deferral against Full Automation and Naive Review

On misclassification cost (Fig. 3), analyst-aware deferral beats full automation with team A at full capacity (−73.00) and with teams B and C at every realistic capacity, and beats naive review with teams A and C.

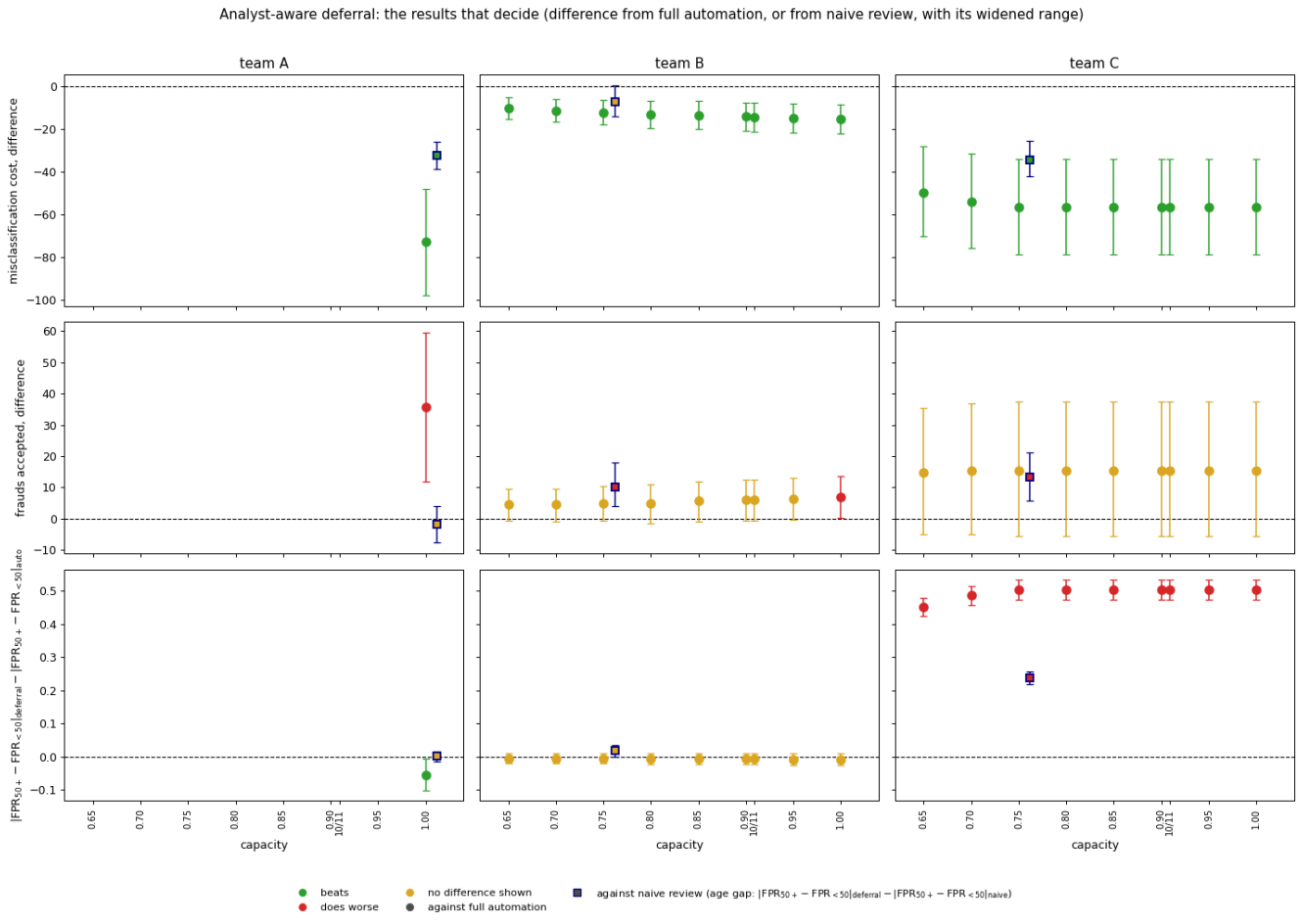


Fig. 3. *Analyst-aware deferral: every deciding result, laid out as Fig. 2, the realistic capacities 0.65 to 1 across; the square marker with a dark blue edge shows the three comparisons with naive review (team A at full capacity, teams B and C at 0.75). Graph: notebook, section 3.2, item P3-F10.*

On frauds accepted, it does worse than full automation with teams A and B at full capacity, and shows no difference elsewhere.

On the age gap, it beats full automation with team A, shows no difference with team B, and does worse with team C at every realistic capacity.

A smaller age gap can come from rejecting more applicants in both groups, so Fig. 4 opens the age-gap verdict at capacity 0.75. With team B, the three policies keep the two age groups about equally far apart; analyst-aware deferral rejects fewer legitimate applicants than the other two in both age groups, and accepts more frauds in both. With team C, both review policies widen the gap mainly by rejecting far fewer legitimate applicants under 50, at the cost of more of their frauds accepted, while applicants aged 50 or over are rejected about as often as under full automation or more often. Under analyst-aware deferral, the wider gap comes almost entirely from the group under 50. In neither team does a policy narrow the gap by rejecting more applicants in both groups. The graph only explains the deciding verdict.

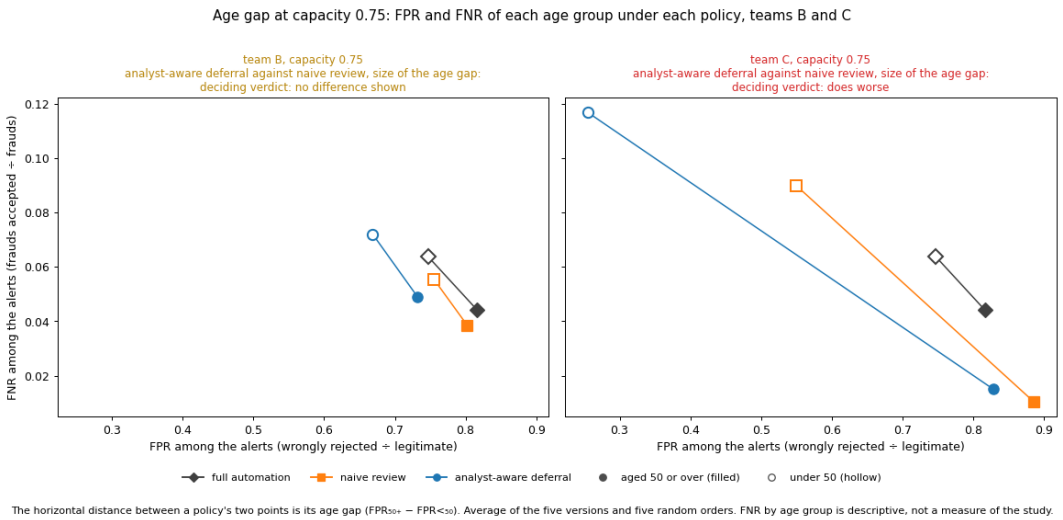


Fig. 4. *FPR and FNR of each age group under each policy, teams B and C at capacity 0.75, among the alerts. Graph: notebook, section 3.2, item P3-F11.*

C. The Verdict, Hypothesis by Hypothesis

Table 2 shows the deciding results against each hypothesis, per team, measure and realistic capacity.

Hypothesis	Description	Compared policies			Team	Misclassification cost									Frauds accepted		Size of the age gap								
		Full automation	Naive HITL	Analyst-aware deferral		0.65	0.70	0.75	0.80	0.85	0.90	10/11	0.95	1.00	0.75	1.00	0.65	0.70	0.75	0.80	0.85	0.90	10/11	0.95	1.00
1	Across realistic review capacity levels, analyst-aware deferral achieves lower risk than full automation	X		X	A									✅		✗									✅
					B	✅	✅	✅	✅	✅	✅	✅	✅	✅	≈	✗	≈	≈	≈	≈	≈	≈	≈	≈	≈
					C	✅	✅	✅	✅	✅	✅	✅	✅	✅	≈	≈	✗	✗	✗	✗	✗	✗	✗	✗	✗
2	Across realistic review capacity levels, analyst-aware deferral achieves lower risk than naive review everything HITL		X	X	A									✅		≈									≈
					B			≈							✗				≈						
					C			✅							✗				✗						
3	Across realistic review capacity levels, full automation can outperform naive HITL when reviewers' capacity is exceeded	X	X		B	≈	≈	≈	≈	≈	≈	≈	✗		≈		✗	≈	≈	≈	≈	≈	≈	≈	
					C	≈	≈	✗	✗	✗	✗	✗	✗		≈		✅	✅	✅	✅	✅	✅	✅	✅	
4	Across realistic review capacity levels, full automation can outperform naive HITL when reviewers largely agree with the model	X	X		B	≈	≈	≈	≈	≈	≈	≈	✗	✗	≈	✅	✗	≈	≈	≈	≈	≈	≈	≈	≈

Table 2. *The hypothesis, per team, measure and realistic capacity. ✅ supports the hypothesis; ✗ refutes it; ≈ no difference shown; a blank cell has no deciding result.*

Analyst-aware deferral lowers misclassification cost against full automation with every team, but it does not lower the number of frauds accepted, which rises with teams A and B at full capacity, and with the biased team C it widens the age gap at every realistic capacity (hypothesis 1). Against naive review it lowers misclassification cost with teams A and C, but accepts more frauds with teams B and C at 0.75 and widens the age gap with team C (hypothesis 2). Exceeded capacity does not make naive review worse than full automation on misclassification cost or frauds accepted; only team C's age gap is wider, and it comes from team C's rule (hypothesis 3). Reviewers who largely agree with the model make naive review worse than full automation only on frauds accepted at full capacity, and the cause is the reviewers' own decisions on the cases they review, not the agreement (hypothesis 4).

The descriptive verdicts at every capacity of the grid are drawn in Figs. C1 to C3 (Appendix C), and every combination is in the full sweep table files (section II-E).

This per-cell reading was set after the results were seen; it summarises the deciding results and changes no verdict.

IV. Descriptive Results

The results in this section decide nothing. Where a cell at a realistic capacity is also a deciding result, the deciding verdict of section III governs.

A. The Baseline

The two automated options swap places at $\lambda \approx 0.046$. Full automation costs $40 + 2{,}878\ \lambda$ (40 frauds accepted, 2,878 legitimate applicants rejected), and confirming every alert costs $3{,}746\ \lambda$, resulting in a difference of 868 legitimate applicants rejected, so the two automated options swap order at $\lambda = 40 \div 868 \approx 0.0461$ [10, section 8.3]. At 0.035 confirming every alert is the cheaper automated option.

Measure	Full automation	Confirm every alert (reference)
Misclassification cost, λ = 0.035	140.73 [128.54, 154.89]	131.11 [129.47, 132.69]
Misclassification cost, λ = 0.0537	194.55 [182.16, 208.75]	201.16 [198.64, 203.58]
Misclassification cost, λ = 0.057	204.05 [191.63, 218.38]	213.52 [210.84, 216.09]
Misclassification cost, λ = 0.171	532.14 [516.73, 549.05]	640.57 [632.53, 648.27]
Frauds accepted	40 [28, 53]	0 (by construction)
FNR	0.0563 [0.0398, 0.0752]	0 (by construction)
Age gap	0.0696 [0.0423, 0.0970]	0 (by construction)
FPR, 50 or over / under 50	0.8166 / 0.7470	1 / 1

Table 3. *Full automation, the baseline, with 95% ranges (1,000 re-draws of the shared index matrix, section II-F) (notebook, section 4.2; table file* `tables/P1-T2_baseline_by_cost_ratio.csv`*)*

B. The Reviewers against Classifier h

This study's first criterion (section IV-E) asks whether the reviewer is more likely than the model to be correct. Before any team rule or routing, every reviewer lies above Classifier *h*'s curve (Fig. 5): on its own, each separates frauds from legitimate applicants better than Classifier *h* could at any cut-off. All but two lie above the dashed line, so they have a lower misclassification cost than Classifier *h* as it is set; the other two lie just below it. The reviewers differ widely in how many legitimate applicants they reject.

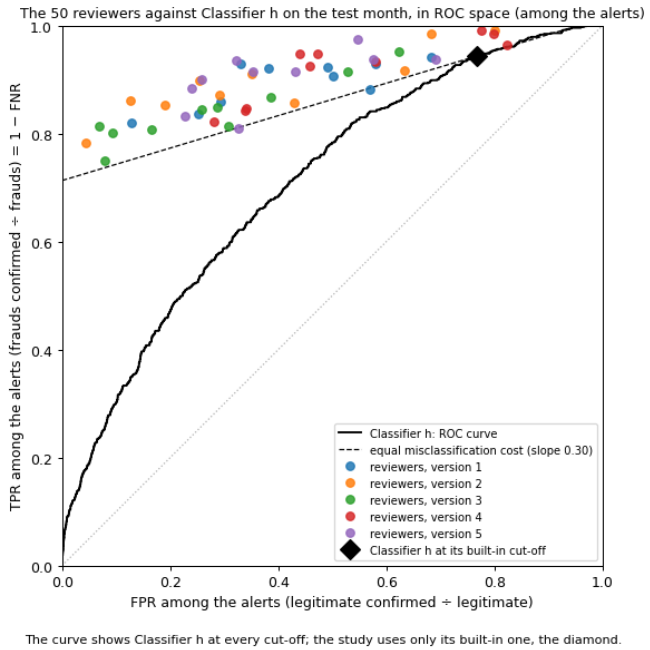
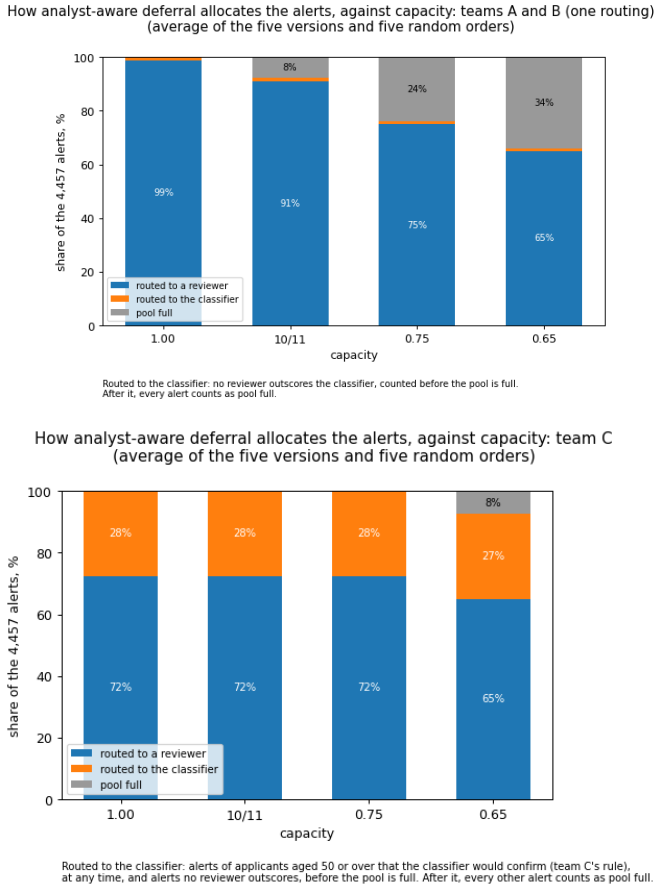


Fig. 5. *The 50 reviewers against Classifier h on the test-month alerts, in ROC space (FPR against TPR, among the alerts): each reviewer is a point, coloured by team version; the solid curve is Classifier h at every cut-off, and the diamond its built-in cut-off; a reviewer above the dashed line of equal misclassification cost through the diamond has a lower misclassification cost than Classifier h as it is set.*

C. Routing, the Router, and Where the Difference against Naive Review Comes From

The notebook splits analyst-aware deferral minus naive review into the alerts left to the classifier and the alerts reviewed, on misclassification cost and frauds accepted. It does so for team A at full capacity and teams B and C at 0.75. The figures are averages of five versions, with 95% ranges (section 4.7, item rows-split; table file `tables/rows-split.csv`). Only the part from the reviewed alerts bears on routing to the right reviewer. In the three deciding comparisons, the advantage on misclassification cost lies in the alerts reviewed. Team B's extra frauds accepted lie in the alerts left to the classifier (+10.28 of +10.16), so they reflect how the alerts not reviewed are treated, not the routing; team C's lie in the alerts reviewed (+13.24 of +13.16). The split does not separate the router's choice of alerts and reviewers from the way the alerts it does not review are treated.

The share decided by the classifier because the pool is full grows as capacity falls (Fig. 6, top). The work is very unevenly shared (Fig. 6, bottom): with teams A and B at full capacity, the two busiest of the ten reviewers take between about two in five and four in five of the reviewed alerts, depending on the version and the random order, and in more than half of the runs at least one reviewer takes none. Which reviewers are favoured changes with the router's training shuffle.



The router's ranking quality is measured as a weighted area under the ROC curve [1] (legitimate applicants weighted 0.057, frauds 1; 0.5 is chance, 1 perfect); the notebook gives the average, lowest and highest of each version's 50 models and of all 250, on the test month beside the previous month (section 4.8, item P3-T1r; table files `tables/P3-T1c_ranking_quality_by_version.csv` and, per model, `tables/P3-T1b_ranking_quality_per_model_month7.csv`).

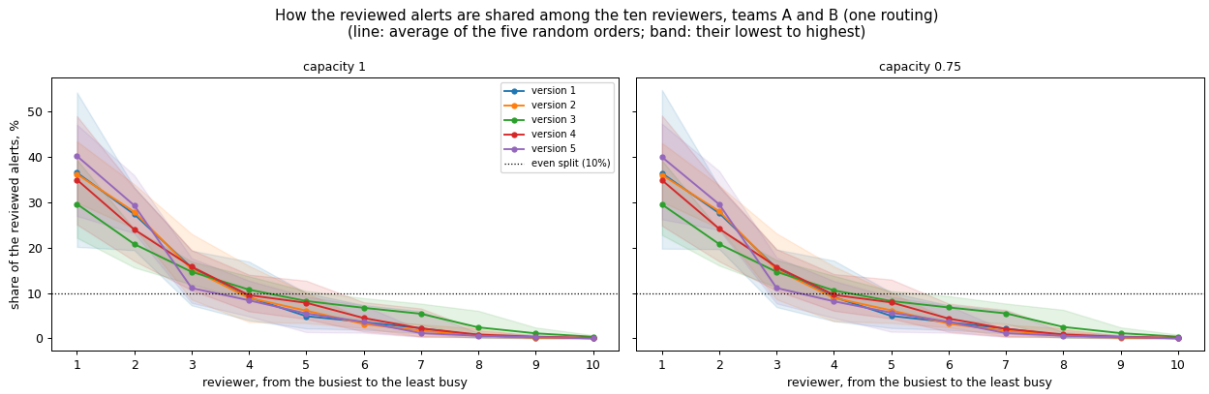


Fig. 6. *Top: share of the alerts routed to a reviewer, routed to the classifier, and decided by the classifier because the pool is full, at capacities 1, 10/11, 0.75 and 0.65 (teams A and B share one routing; team C has its own). Bottom: each reviewer's share of the reviewed alerts, teams A and B at capacities 1 and 0.75, busiest first. Graphs: notebook, section 4.8, item P3-T1.*

Averaged over each version's models, the router ranks only a little better than chance. **The router is weak:** the router trained here ranks the reviewers' correctness only weakly, so the results on analyst-aware deferral hold for this router; they do not show what a better router would give.

D. Access to a Reviewer by Age Group, and Team C's Bias against Capacity

Under naive review, both age groups reach a reviewer in proportion to capacity (Fig. 7). Under analyst-aware deferral with team C, from a capacity of 0.75 upwards, applicants aged 50 or over reach a reviewer only about one time in six, while applicants under 50 almost always do; below 0.75 both shares are lower, and the older group's never exceeds 16%. None of the older applicants whom the classifier would reject ever reaches a reviewer: team C's rule would reject them whatever the reviewer decided (behaviour diagram), so the routing sends them straight to the classifier (Appendix A-E). With a biased team, analyst-aware deferral quietly denies most older applicants a human review. At capacity 0.75, analyst-aware deferral with team C sends 15.65% of the alerts of applicants aged 50 or over to a reviewer, and 98.95% of those of applicants under 50 (table file `tables/P3-T9_access_by_age.csv`).

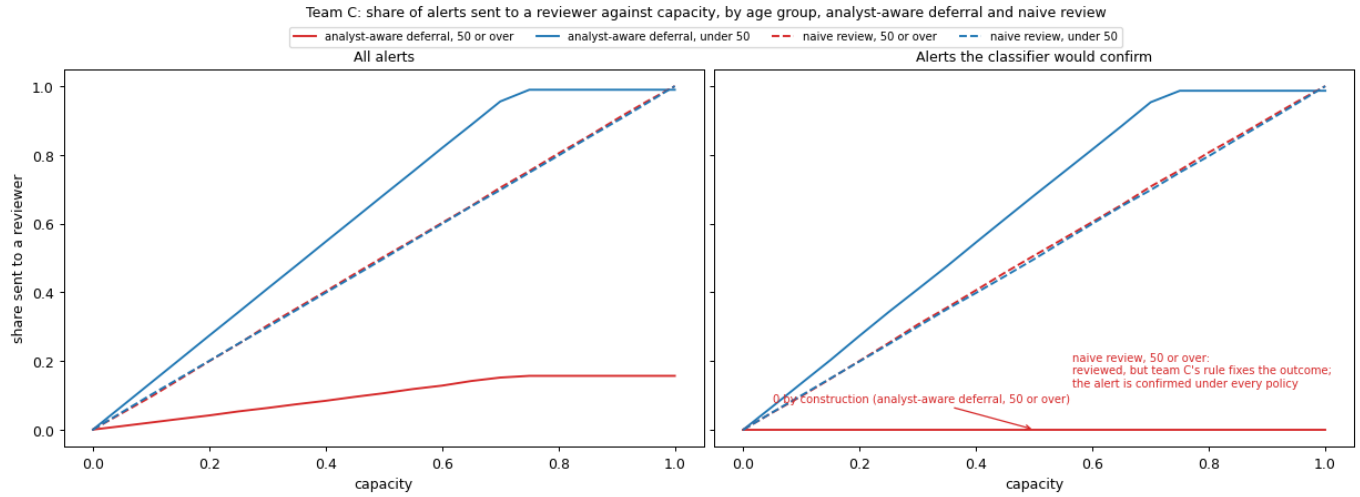


Fig. 7. *Team C: share of alerts sent to a reviewer against capacity, by age group; solid lines analyst-aware deferral, dashed lines naive review; red aged 50 or over, blue under 50. For teams A and B both groups lie on naive review's diagonal and are not drawn. Graph: notebook, section 4.5, item P3-F9.*

Over every application of the test month, counting the applications the Alert Model did not flag as accepted, the FPRs are far lower than among the alerts (Table 4). Within each age group the policies keep the same order as among the alerts, because each group's denominator is fixed.

Policy	Team, capacity	50 or over	Under 50	All
Full automation	—	0.0763	0.0234	0.0302
Naive review	A, 1	0.0384	0.0124	0.0158
Naive review	B, 0.75	0.0750	0.0236	0.0302
Naive review	C, 0.75	0.0828	0.0172	0.0256
Analyst-aware deferral	A, 1	0.0251	0.0080	0.0102
Analyst-aware deferral	B, 0.75	0.0684	0.0209	0.0270
Analyst-aware deferral	C, 0.75	0.0775	0.0080	0.0169

Table 4. *FPR over every application of the test month, at the deciding comparisons; average of five versions and five random orders; descriptive (notebook, section 4.5, item P3-T10)*

This study also asks whether a bias compounds or cancels when capacity is exceeded; the policy 2 design sets out how this is read [10, section 9]. At capacity 0 every application among the alerts is rejected, so both age groups start at an FPR of 1 (Fig. 8). As capacity rises, team A lowers both groups' FPR together; team C lowers that of applicants under 50 in the same way, but that of applicants aged 50 or over much less. At full capacity, team C's age gap is about 0.44 larger than team A's. **Overload does not make team C's bias worse; it hides it.** Team C's excess over team A shrinks in proportion to capacity. **The one finding is on misclassification cost, and it is small:** team C's extra cost over team A is 2.2 lower at 0.75 than at full capacity (0.2 to 4.0), read as "cancels (in part)"; only three of the five versions show it, and the other two read "additive".

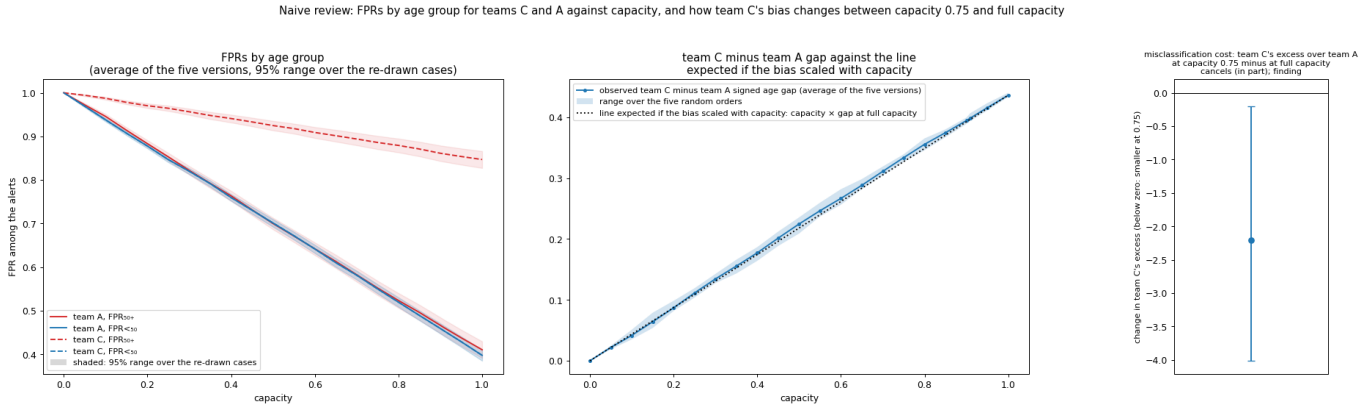


Fig. 8. Naive review: FPR by age group for teams C and A against capacity, and, in three small panels, team C's excess over team A at 0.75 minus the same at full capacity, on the signed age gap, the wrongly rejected applicants aged 50 or over, and misclassification cost. Graph: notebook, section 4.6, item P2-F5.

E. Monitoring Indicators and the Four Criteria

This study proposes four criteria that make the outline's objective measurable: a case should reach a human only if:

1. the reviewer is more likely than the model to be correct;
   Criterion 1 is built into analyst-aware deferral as the router sees it, since a reviewer takes an alert only if the router's estimate is strictly above the classifier's confidence, and whether it holds in fact is measured.
2. the volume routed fits the actual review capacity;
   Criterion 2 is built in: never more alerts are reviewed than capacity allows.
3. the misclassification cost avoided exceeds the cost of review;
   Criterion 3 is measured, not enforced: no cost of review is modelled, and the cost avoided per review is shown as the cost of one review at which review breaks even, in units of a missed fraud.
4. the reviewer has the information and time to disagree with the model.
   Criterion 4 cannot be observed directly in a simulation without time pressure.

The outline proposes four indicators a bank could compute from its own review logs [4]: reviewer override rate, capacity utilisation, "loss avoided per review", called cost avoided per review here, and subgroup disparity. Under naive review every review place is always used, so capacity utilisation is shown as alerts per review place (1 ÷ capacity, the same for every team); under analyst-aware deferral, as the share of review places used (Fig. 9). The indicators are not defined at capacity 0.

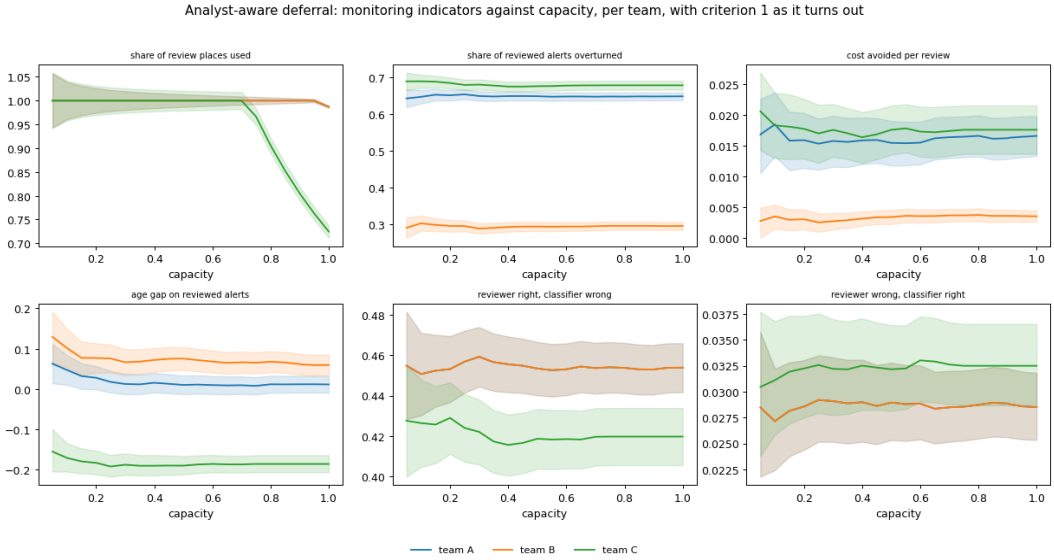


Fig. 9. Analyst-aware deferral: the four indicators per team against capacity, with two panels showing criterion 1 as it turns out (on reviewed alerts, the share where the reviewer is right and full automation wrong, and the reverse). Graph: notebook, section 4.3, item P3-F5.

V. How to Apply the Study in the Real World

A. What Would Stay Relevant, and What Would Need to Change

Table 5 lists author's view of the assets of a real-world design option, what each component does, the study assets that fit it with no change or a minimal change, and the changes those assets would require.

#	Real-world asset	What it does	Study assets	Changes Required
1	Alert-RW model	Scores every application. The higher the score, the higher the probability of fraud	The study's Alert Model, which scores every application	Add the study's weights (1 for fraud, 0.057 for legitimate). Select the model with the lowest FPR under the 1% FNR cap, instead of the best catch rate at 5% FPR
2	Alert-RW threshold	Splits the scored applications: below it, an application is accepted; at or above it, it becomes a Flag	The study's cut-off calculation for the Alert Model	Apply the same ranking rule to frauds, to meet the 1% FNR cap instead of 5% FPR
3	Flags	The applications at or above the Alert-RW threshold, the only ones that can reach a reviewer	The study's step that keeps the applications above the cut-off, with their scores	Use the cut-off from row 2, and keep applications scoring at or above it (the study keeps only those strictly above it)
4	Fair-fairy model	Scores the Flags only, to pick out legitimate applicants who can be accepted without review and so reduce fairness exposure. It is judged on the gap between groups	The study's Classifier *h*, fitted only on alerts with the weighted loss	No change
5	Fair-fairy threshold	Splits the Flags: below it, a Flag goes to a reviewer; at or above it, the application is accepted without review	—	—
6	Router-New, option 1	The fixed rule that applies the two thresholds and sends each application either to acceptance without review or to a reviewer	The study's naive review policy, with team A's rule (the reviewer's decision stands)	Add one test before a Flag goes to a reviewer: Fair-fairy score below its threshold. Drop the overflow rule, because capacity is unlimited
7	Team of experts (the reviewers)	Decide the Flags sent to them, labelling each as fraud or legitimate without seeing the models' scores. Their labels train the models	—	—
8	Labelled training data	The applications the experts have labelled, over X months (assumed 3), used to train the models	The study's builder of the reviewers' training table (one reviewer decision per case), including its fix to team membership	Use the real assignments and decisions from the logs instead of simulated ones; Flags only
9	Flag and threshold drift detection solution	Compares the reviewers' decisions with the models' scores to support the regular threshold adjustments (for example quarterly), and detects drift for the worse that triggers a re-fit of Alert-RW	—	—
10	Second line of defence team	Senior experts who review a random sample of the reviewers' decisions and classify each as correct or incorrect	—	—
11	Reviewer effectiveness scoring solution	Scores each reviewer from the second line's verdicts. It is deterministic	The study's per-reviewer error rates, computed from decisions against labels	Use the second line's verdict (correct or incorrect) instead of the dataset's fraud label

Table 5. *The assets of a real-world design: what each does, the study assets it re-uses, and the changes they require*

B. Further Aspects for Commercial Use

The project is academic only and nothing here confirms compliance. The study is a generic HITL framework leveraging fraud detection data. Therefore, the anti-money-laundering regulation has not been assessed and must be done for real-world use. The following would also need to be worked out:

Applications accepted with no human involved. Every decision taken with no person involved is an acceptance. Every rejection is made by a reviewer who does not see the scores. Accepting an application with no human involved is an automated decision, but one that gives the applicant what they asked for. Four things would still need working out:

1. **AI literacy.** Staff who run or use the models need enough AI literacy. Both models detect fraud, so neither is high-risk credit scoring, and this is the only AI Act duty.
2. **EU safeguards still apply.** Solely automated decisions are allowed where they are necessary to enter into a contract with the applicant. It would have to be shown that deciding with no person is actually necessary, not just convenient. That opening the account counts as such a contract is an assumption to confirm. Even then, the bank needs "suitable measures": at least the right to ask for a person, to give a view and to contest. Applicants must be told that automated decisions are made, how the logic works and what follows for them.
3. **Swiss exemption, two conditions.** The duty to inform and the right to be heard are lifted only if the decision is directly connected with concluding or performing a contract with the applicant, *and* the applicant's request is granted. Both must hold.
4. **Access by age.** The Fair-fairy threshold decides who is accepted with no person and who goes to a reviewer, and only a reviewer can reject. The impact assessment would need to check the share accepted with no person, by age group, for discriminatory effects.

Records. A record for each decision of whether a person decided or the models' outcome stood, to answer access requests.

Second line of defence. If the reviewers' effectiveness scores are computed by machine, even by a fixed rule, that is profiling of their performance at work. If the senior experts compile them by hand, it is not. (A fixed rule written by people is not an AI system, so the AI Act's high-risk check falls away.)

Learning loop. A re-fit or a threshold change leaves what past applicants were told correct. New applicants are told the logic in force when they apply. The records need to show which version decided each application. The impact assessment needs a review only when a change alters the risk. Changing thresholds to meet the service level agreement changes the share accepted with no person, and possibly the age gap. That is a change of risk, which calls for a review of the impact assessment. Reviewers' errors, including any age bias, enter the training targets, so the age gap needs re-checking after each re-fit.

Labels the second line finds incorrect. The accuracy principle requires inaccurate personal data to be corrected or erased promptly. Using the corrected label in the next retraining puts the training data right. Three things would still need settling: whether the label on the applicant's record is corrected when the error is found, rather than only at the quarterly retraining; whether the applicant's own decision is reopened, which matters more if "incorrect" includes wrongly rejected applicants; the Swiss equivalent, which needs a legal assessment.

VI. Discussion

A. Departures from the Project Outline

Policies. Analyst-aware deferral, one of the outline's aspirational extensions, replaces calibrated deferral, as it better illustrates the objective; the hypothesis and the first refutation condition are tested with it.

Reviewers and teams. The decisions replayed are those of 50 reviewers regenerated with the dataset's generator, without weight on age and fitted on four months, not the dataset's recorded decisions. The age weight was removed so that the teams are alike except for team C, whose age bias is built into its behaviour. Fitting on four months sets reviewers' baseline settings on more alerts than one month provides.

B. Limitations

The reviewers are simulated. Each makes one fixed decision per alert; there is no fatigue, time pressure or learning, and exceeding capacity changes only what happens to the alerts beyond it. This study's criterion 4 (section IV-E) cannot be observed directly.

No cost of review is modelled. Cost avoided per review is shown as the break-even cost instead.

One alert model, one test month: 4,457 alerts, 711 of them frauds. Frauds the Alert Model did not flag are outside the study.

The cost ratio is implied, not observed. The data suggest a ratio of roughly 0.03 to 0.045; of that range only 0.035 is tested. Away from 0.057, neither full automation, the reviewers nor the router is re-tuned, and how much better a re-tuned automation would do is not measured.

Team C's bias is a rule, not a model of how bias arises in people.

The router is weak. It ranks the reviewers' correctness only weakly. A router that fits the reviewers less well puts analyst-aware deferral at a disadvantage.

The routing's own effect is not separated from how the alerts not reviewed are treated (section IV-C). When analyst-aware deferral beats or loses to naive review, we can't tell how much of that comes from routing alerts to the right reviewer and how much from what happens to the alerts nobody reviews.

Deferral capacity is modelled as one shared pool. The router sends most of the work to a few reviewers (section IV-C). No real team could absorb such a split, and fatigue

is not modelled, so analyst-aware deferral is shown under conditions more favourable than a real team would offer.

The ranges cover which cases fell in the test month, not the policy's own randomness, which is shown as the range over random orders. Judging on all five versions or on their average (section II-F) makes the 5% chance of a false finding approximate. The deciding results of the two policies are adjusted separately, so across both together that chance is above 5%.

The project outline's own limitation on the author's dual role, as author and as stakeholder, stands.

C. Risks

The risks named in the project outline were mitigated by the author as part of the project.

VII. Conclusion

Each measure is judged on its own, never combined.

Misclassification cost. The hypothesis is refuted by the second condition: naive review never falls below full automation at the deciding capacities. Analyst-aware deferral beats full automation for every team at every realistic capacity at which it is judged, but is not shown to beat naive review in all three comparisons.

Frauds accepted. The hypothesis is refuted by the first condition: analyst-aware deferral does not beat full automation for every team and capacity. Naive review can fall below automation on this measure, through the reviewers' own decisions on the alerts they review (teams A and B at full capacity); the rule for reviewers who agree with the model is supported here, but not through the agreement.

Size of the age gap. The hypothesis is refuted by the first condition. Naive review falls below automation through team C, whose only difference from team A is its bias rule; overload dilutes that bias rather than accentuating it.

On each measure, as the project outline states, "the benefit of human review is smaller and **more conditional than assumed**" [4]. The first refutation condition is read strictly: one deciding result in which analyst-aware deferral fails to beat full automation refutes the hypothesis on that measure. Only the age-gap verdict depends on this reading; on misclassification cost and frauds accepted, the refutation holds under any reading. Table 2 shows where each policy works, capacity by capacity.

References

[1] J. V. Alves, D. Leitão, S. Jesus, M. O. P. Sampaio, J. Liébana, P. Saleiro, M. A. T. Figueiredo and P. Bizarro, "A benchmarking framework and dataset for learning to defer in human-AI decision-making," *Scientific Data*, vol. 12, art. 506, 2025, *https://doi.org/10.1038/s41597-025-04664-y*. It numbers the months from 1, so its month 8 is month 7 of the data files [2]. Licensed under CC BY-NC-ND 4.0; quoted with citation, no figure reproduced.

[2] Financial Fraud Alert Review Dataset (FiFAR), version 1, Alves et al., figshare, 2025, digital object identifier 10.6084/m9.figshare.28351172, *https://doi.org/10.6084/m9.figshare.28351172*, licensed under CC BY 4.0, *https://creativecommons.org/licenses/by/4.0/*. Its `Base.csv` derives from the Bank Account Fraud dataset, credited in the repository README [5], section "Data: FiFAR".

[3] OpenL2D, *https://github.com/feedzai/openl2d/tree/main*, Apache License 2.0.

[4] Z. Sipka, "Save the Human (in the loop): when does human oversight of AI actually reduce risk?", Innovation Project Outline, CAS Machine Learning in Finance and Insurance, ETH Zurich, version 0.9, 18 September 2026.

[5] SaveHITL-pub, *https://github.com/zsipka/SaveHITL-pub*: `savehitl_results_final.ipynb` (the notebook); the table files are written under `tables/` when the notebook is run.

[6] Regulation (EU) 2024/1689 (AI Act), Article 2(6) and (8), Article 4 and Annex III, point 5(b); Article 4 as replaced by Regulation (EU) 2026/1744 (adopted text PE-CONS 30/26), Article 1, point (5).

[7] Regulation (EU) 2016/679 (General Data Protection Regulation), Articles 4(4), 5(1)(d), 13(2)(f), 15(1)(h), 22, 35(1), (3)(a) and (11), and recital 71; Commission proposal COM(2025) 837.

[8] Swiss Federal Act on Data Protection, Articles 5(f), 21, 22 and 25(2)(f) (English translation, not binding).

[9] "Policy 1: Full automation (the baseline)," policy design, version 1.0, SaveHITL-pub, `design/policy-1-design_v1.0.md`.

[10] "Policy 2: Naive HITL ("review everything")," policy design, version 1.0, SaveHITL-pub, `design/policy-2-design_v1.0.md`.

[11] "Policy 3: Analyst-aware deferral," policy design, version 1.0, SaveHITL-pub, `design/policy-3-design_v1.0.md`.

Appendix

A. Detailed Deciding Results

Means of the five versions; λ = 0.057. Adjusted ranges are given for the single comparisons; for the per-capacity results, the verdict is marked (**b** beats, **w** does worse, no mark: no difference shown), and the ranges are drawn in Figs. 2 and 3.

Comparison, measure	Estimate [adjusted range]	Verdict
A at 1 − automation, misclassification cost	−40.76 [−65.94, −15.01]	beats
A at 1 − automation, frauds accepted	+37.52 [13.20, 62.26]	does worse (reviewed alerts)
A at 1 − automation, size of age gap	−0.0567 [−0.1014, −0.0061]	beats
B − A at 0.75, misclassification cost	+22.83 [6.50, 38.76]	does worse
B − A at 0.75, frauds accepted	−23.96 [−39.98, −8.44]	beats
B − A at 0.75, size of age gap	+0.0428 [−0.0099, 0.0698]	no difference shown
C − A at 0.75, misclassification cost	+5.31 [−5.23, 14.34]	no difference shown
C − A at 0.75, frauds accepted; signed age gap	−16.48; +0.3338	follows from team C's rule

Table A1. *Naive review: team A at full capacity against full automation, and teams B and C against team A at 0.75 (table file* `tables/P2-T3_deciding_results.csv`*, written by section 3.1 of the notebook; drawn in Fig. 2*

Capacity	B: cost	B: frauds	B: gap	C: cost	C: frauds	C: gap
0.65	−3.11	−10.24	−0.0289 **b** (a)	−17.39	−2.68	+0.2217 **w**
0.70	−3.47	−7.28	−0.0254 (a)	−19.31	+0.16	+0.2428 **w**
0.75	−4.89	−5.24	−0.0228 (a)	−22.42 **b**	+2.24	+0.2663 **w**
0.80	−5.55	−2.56	−0.0177	−24.37 **b**	+5.24	+0.2900 **w**
0.85	−6.83	−0.64	−0.0145	−26.21 **b**	+8.44	+0.3122 **w**
0.90	−7.83	+1.60	−0.0148	−28.48 **b**	+11.24	+0.3323 **w**
10/11	−8.03	+2.04	−0.0144	−29.25 **b**	+11.40	+0.3366 **w**
0.95	−8.30 **b**	+4.40	−0.0122	−30.56 **b**	+14.16	+0.3553 **w**
1.00	−8.09 **b**	+7.84 **w** (b)	−0.0079	−33.25 **b**	+16.56	+0.3796 **w**

Table A2. *Naive review − full automation, teams B and C, realistic capacities (table files* `tables/P2-T3_deciding_results.csv` *and* `tables/P2-T3b_realistic_range_verdicts.csv`*, written by section 3.1 of the notebook; drawn in Fig. 2). (a) Labelled "gap narrowed by rejecting more applicants in both groups"; it does not show that team B treats the two groups more fairly. (b) The reviewers themselves do worse than full automation on the alerts they review; the unreviewed part is 0.*

Comparison, measure	Estimate [adjusted range]	Verdict
A at 1 − automation, misclassification cost	−73.00 [−98.06, −48.27]	beats
A at 1 − automation, frauds accepted	+35.80 [11.87, 59.52]	does worse
A at 1 − automation, size of age gap	−0.0559 [−0.1014, −0.0055]	beats
A at 1 − naive review, misclassification cost	−32.23 [−38.63, −25.99]	beats
A at 1 − naive review, frauds accepted	−1.72 [−7.72, 4.04]	no difference shown
A at 1 − naive review, size of age gap	+0.0008 [−0.0154, 0.0145]	no difference shown
B at 0.75 − naive review, misclassification cost	−7.29 [−13.92, 0.53]	no difference shown
B at 0.75 − naive review, frauds accepted	+10.16 [3.88, 17.80]	does worse
B at 0.75 − naive review, size of age gap	+0.0166 [−0.0001, 0.0332]	no difference shown (c)
C at 0.75 − naive review, misclassification cost	−34.41 [−42.41, −25.53]	beats
C at 0.75 − naive review, frauds accepted	+13.16 [5.76, 21.24]	does worse
C at 0.75 − naive review, size of age gap	+0.2381 [0.2186, 0.2567]	does worse (c)

Table A3. *Analyst-aware deferral: team A at full capacity against full automation, and the three comparisons with naive review (table file* `tables/P3-T3_deciding_results.csv`*, written by section 3.2 of the notebook; drawn in Fig. 3)*

(c) Naive review's side labelled "gap narrowed by rejecting more applicants in both groups".

Capacity	B: cost	B: frauds	B: gap	C: cost	C: frauds	C: gap
0.65	−10.23 **b**	+4.48	−0.0058	−49.80 **b**	+14.68	+0.4507 **w**
0.70	−11.35 **b**	+4.52	−0.0062	−54.25 **b**	+15.44	+0.4867 **w**
0.75	−12.18 **b**	+4.92	−0.0062	−56.83 **b**	+15.40	+0.5044 **w**
0.80	−13.22 **b**	+4.92	−0.0069	as 0.75	as 0.75	as 0.75
0.85	−13.45 **b**	+5.68	−0.0063	as 0.75	as 0.75	as 0.75
0.90	−14.27 **b**	+5.96	−0.0073	as 0.75	as 0.75	as 0.75
10/11	−14.45 **b**	+6.00	−0.0075	as 0.75	as 0.75	as 0.75

Capacity	B: cost	B: frauds	B: gap	C: cost	C: frauds	C: gap
0.95	−14.94 **b**	+6.44	−0.0093	as 0.75	as 0.75	as 0.75
1.00	−15.32 **b**	+6.92 **w**	−0.0094	as 0.75	as 0.75	as 0.75

Table A4. *Analyst-aware deferral − full automation, teams B and C, realistic capacities (table files* `tables/P3-T3_deciding_results.csv` *and* `tables/P3-T3b_realistic_range_verdicts.csv`*, written by section 3.2 of the notebook; drawn in Fig. 3). Team C's results are identical from 0.75 to 1: its number of alerts eligible for a reviewer (3,131 to 3,258 per version and random order) is below the capacity at 0.75 (3,342 alerts) in every routing, so its routing no longer changes with capacity (table file* `tables/P3-T1a_routing.csv`*, team C's routing).*

A-E. Outcomes predicted from the rules, and how they turned out

Team B moves only about a fifth of the way from full automation. Held: at full capacity, team B's difference is 0.20 of team A's on misclassification cost and 0.21 on frauds accepted under naive review; 0.21 and 0.19 under analyst-aware deferral (from Tables A1 to A4; for example 7.84 ÷ 37.52 = 0.21).

Under naive review, the alerts beyond capacity raise misclassification cost at every realistic capacity below full (point value). Held at 0.65 to 10/11; not held at 0.95 (−0.15). No range claim was made (the part from the alerts beyond capacity, teams B and C, in `tables/P2-T3_deciding_results.csv`).

Team B against team A at 0.75 takes the opposite direction to team A against full automation. Held on misclassification cost and frauds accepted; on the age gap no difference is shown (Table A1).

Under naive review, team C accepts no more frauds than team A, and its signed age gap is at least team A's. Held in every random order of every version at all 22 capacities (team C minus team A, in `tables/P2-T5_full_sweep.csv`).

On the applications naive review rejects without review, analyst-aware deferral accepts as many frauds or more and has the lower misclassification cost. Held for teams B and C at capacity 0.75 in the mean of five versions (frauds accepted +10.84 and +14.92; misclassification cost −1.93 and −15.99); single random orders can differ (table file `tables/P3-T8_point_figures.csv`).

Under analyst-aware deferral, team C sends no applicant aged 50 or over whom Classifier h would reject to a reviewer, and team B's access figures equal team A's. Held in every version, random order and capacity: team C's figure is 0 by the routing rule (section IV-D), and teams A and B share one routing (Fig. 6); every team's shares at every capacity are in `tables/P3-F9_lines.csv`.

Appendix B. The Data by Age Group

In every month, fraud is more frequent among applicants aged 50 or over than among younger applicants (Fig. B1), and in point value its rate rises from one age group to the next. The Alert Model flags a larger share of older legitimate applicants. Among the features, `date_of_birth_distinct_emails_4w`, `phone_home_valid`, `credit_risk_score` and `current_address_months_count` are the most correlated with age, so they too carry information about age. Panel D4 leaves out the categorical features, whose values have no order to rank (`payment_type`, `employment_status`, `housing_status`, `source`, `device_os`); `device_fraud_count`, which has the same value for every application; and `month`, the time index. The figure decides nothing.

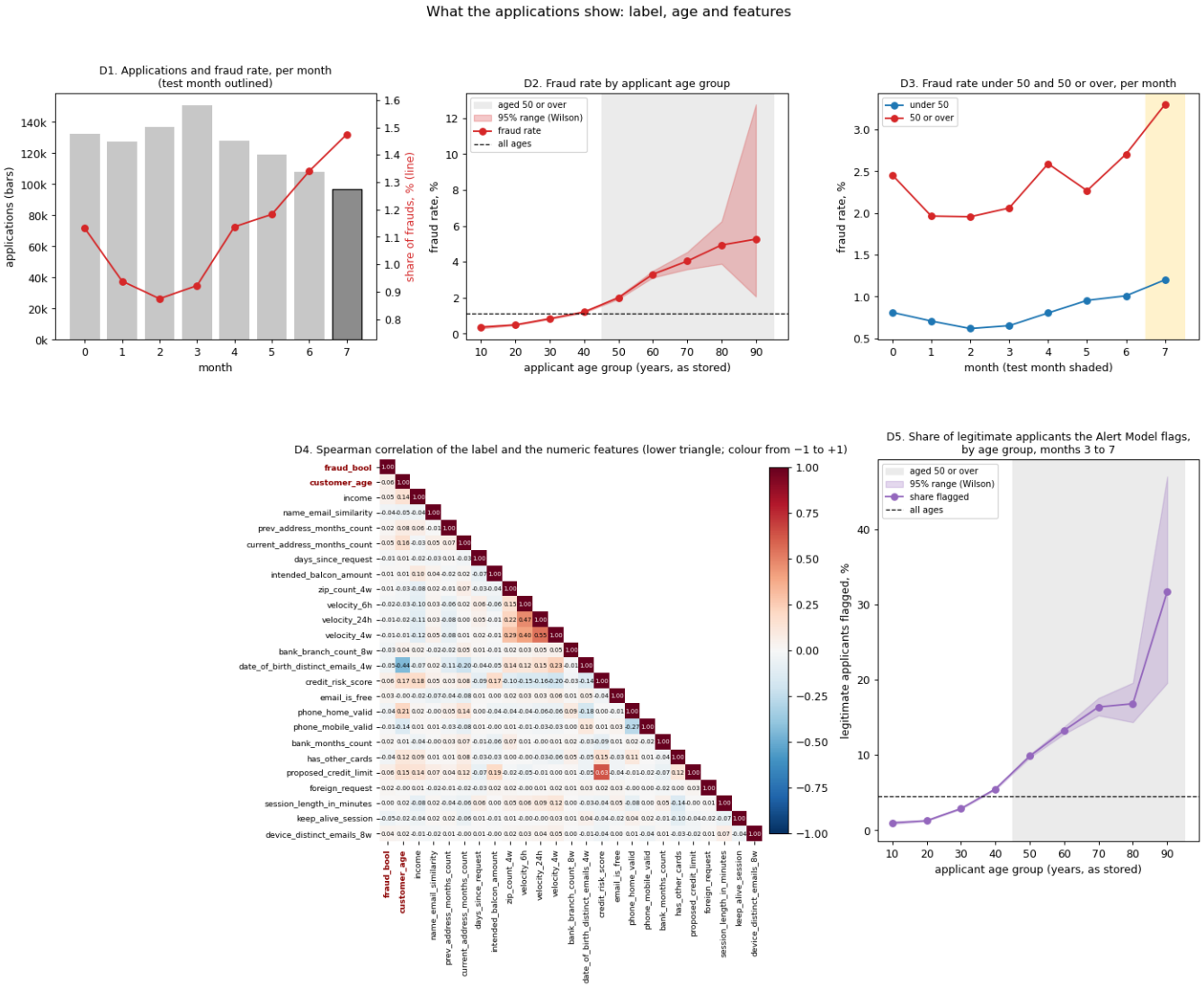


Fig. B1. *What the applications of `Base.csv` (1,000,000 applications) show, in aggregate: D1, applications and fraud rate per month, test month outlined; D2, fraud rate by age group; D3, fraud rate under 50 and at 50 or over, per month; D4, Spearman correlation of the label and the numeric features, with the label and age highlighted; D5, share of legitimate applicants the Alert Model flags, by age group, months 3 to 7. Graph: notebook, section 2, item F-data. These figures summarise FiFAR's `Base.csv` in aggregate: rates, counts and correlations over groups of at least 10 applications. They show no individual application or value.*

Appendix C. Descriptive Verdicts across the Capacity Grid

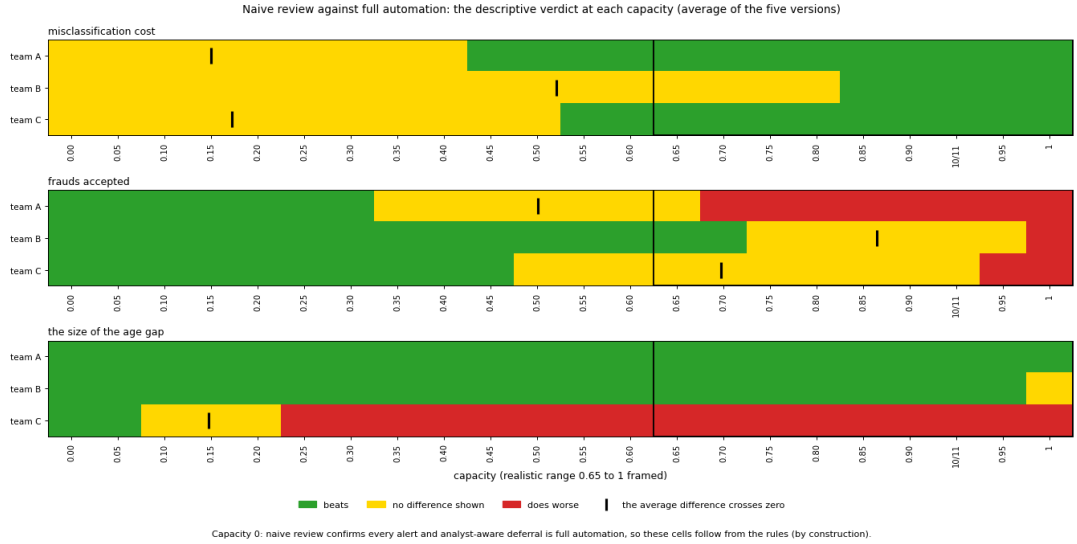


Fig. C1. *Naive review against full automation: descriptive verdict per team and measure at every capacity of the grid, with a tick where the average difference crosses zero. Graph: notebook, section 4.1, item P2-T4.*

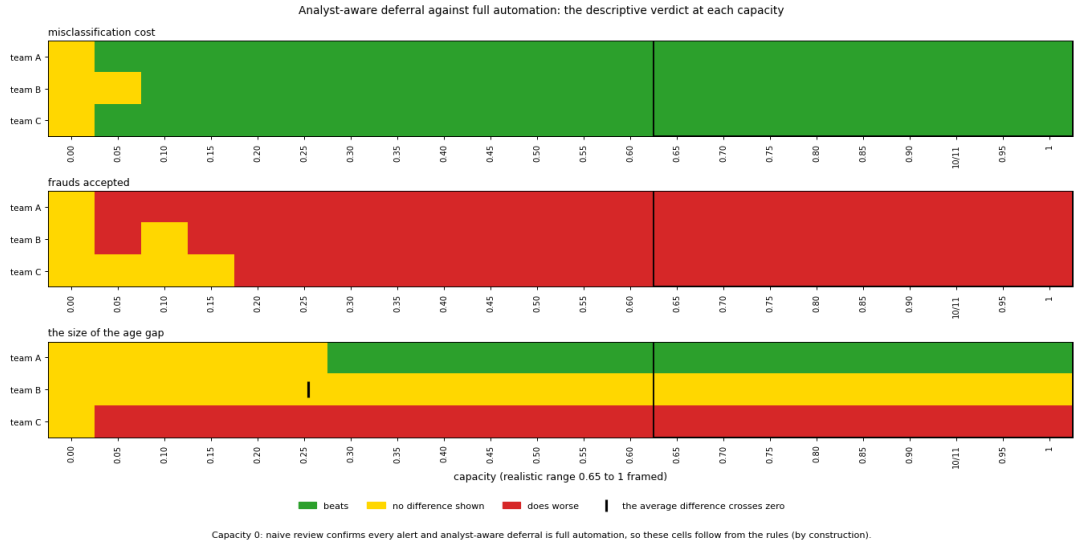


Fig. C2. *Analyst-aware deferral against full automation: descriptive verdict per team and measure at every capacity. Graph: notebook, section 4.1, item P3-T4.*

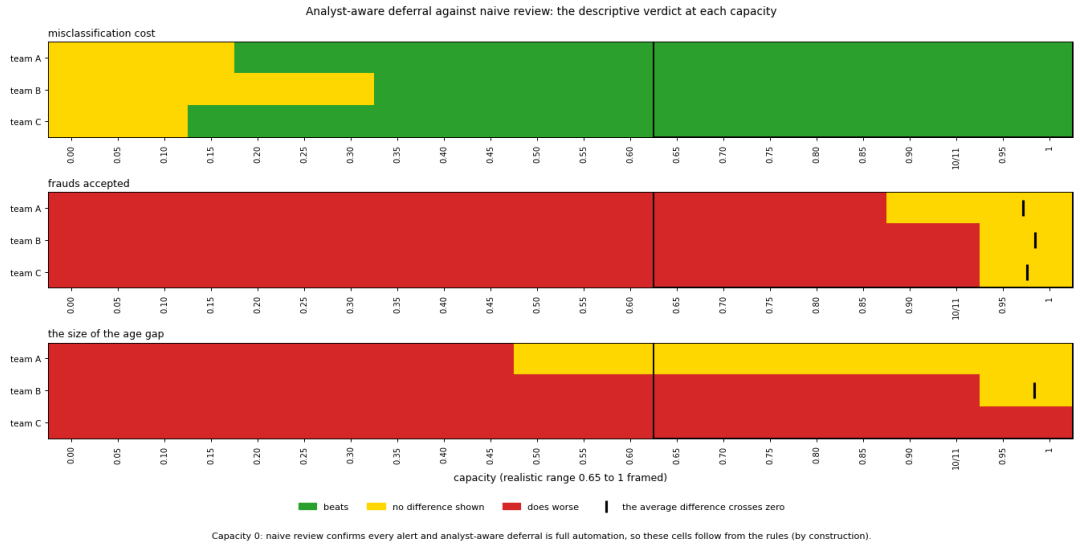


Fig. C3. *Analyst-aware deferral against naive review: descriptive verdict per team and measure at every capacity. Graph: notebook, section 4.1, item P3-T4.*